Induced Acoustic Resonance for Noninvasive Bone Fracture Detection Using Digital Signal Processing and Machine Learning

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***Abstract*—A bone fracture is a complete or incomplete discontinuity in a bone, often caused by an impact. While extreme fractures are sometimes obvious, most fractures require radiographic imaging (such as X-ray) to diagnose and treat. Unfortunately, cost, access to such equipment, and availability of trained personnel to interpret the results present significant barriers to many in remote areas and developing countries. In this feasibility study, a low-cost and portable bone fracture detection method and device are proposed to help this underserved segment of patients. Drawing on previously published work regarding the automated detection of mechanical fractures using induced vibrations in an industrial setting, this paper presents a technique to replicate and improve upon manual detection techniques using a tuning fork and stethoscope by using digital signal processing and machine learning techniques. In order to make fracture detection more accessible, the prototype device presented does not require any specialized skills to operate, maintains portability, is automated, and has the potential to be manufactured inexpensively. Fractures are detected by inducing vibrations in the bone and measuring the resulting signal to detect structural defects. Using animal bones with synthetic soft tissues to replicate the dampening effects of muscle and connective tissue, machine learning models were trained and tested, achieving 93.6% accuracy. The proposed technique may also prove effective in-vivo although further testing is required.**

***Index Terms*—Acoustic resonance, bone fracture, fracture detection, digital signal processing (DSP), embedded systems, machine learning (ML), medical screening.**

I. Introduction

Bone fractures are a complete or incomplete discontinuity in a bone [1]. They are often caused by impact with another object but can also be caused by accumulated stress, osteoporosis, or other medical conditions [2]. Untreated fractures can result in decreased mobility, pain, deformity, difficulty performing normal activities, anxiety, and depression [3].

Bone fractures contribute largely to global years lived with disability (YLD) [4]. Researchers found that femur fractures went untreated in up to 40% of cases (depending on region) and that untreated fractures can increase YLD by 72% [5]. Additionally, many developing healthcare systems do not focus on emergency care [6].

Bone fractures typically require radiographic imaging (e.g., X-ray) to diagnose and correctly treat. However, access to radiographic services and their cost may be barriers to treatment for many, particularly for those living in remote and/or low resource areas. In Sub-Saharan Africa, for example, several countries were served by fewer than 30 imaging personnel [7]. Moreover, the bulk of imaging staff were radiographers, trained to operate, but not interpret, medical imaging results [7].

This paper contributes a low-cost, portable bone fracture detection method using acoustic signal processing and machine learning. A prototype system was implemented and evaluated on its performance. The proposed method can classify a bone fracture with 93.6% accuracy.

II. Related Work

Apart from well-known methods to diagnose fracture, researchers have explored many techniques including ultrasound [8] and tuning forks [9]–[11]. The method utilizing tuning forks has been widely explored for a variety of applications such as augmenting non-radiographic tests in athletic and clinical settings [9], long bone [10] fracture diagnosis, and ankle inversion diagnosis [11]. In all current medical implementations, the tuning fork method requires a trained

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medical practitioner listening through a stethoscope to make a diagnosis.

Although Digital Signal Processing (DSP) and Machine Learning (ML) for acoustic analysis have not yet been used for detecting fracture in bone, these methods have been widely explored in other domains. For example, acoustic diagnostics [12] is a proposed method for material crack detection in applications such as industrial machinery, structural components in buildings, and eggshells [13]–[15]. In each application, an arbitrary acoustic signal is induced into one end of the material, and the resulting signal at the other end is analyzed.

Thus, the tuning fork method has been verified by medical researchers. Existing research also demonstrates the feasibility of using ML and DSP in detecting fractures in other materials such as eggshells and machinery. Automated, lower cost fracture detection may be beneficial for initial fracture screening in low-resource areas, or where quick diagnoses are required.

III. Methods

An acoustic signal induced in bone will be predictably modulated by discontinuity in that bone. Physicians are able to use the tuning fork method to diagnose fracture with up to 81% accuracy [13]. However, the tuning fork method relies on comparison between an unbroken and broken bone to diagnose a fracture and requires trained personnel to execute effectively [13]. In order to improve upon the tuning fork method and make it more accessible, the proposed technique replaces the human ear with a microphone, the tuning fork with a piezoelectric transducer, and trained personnel with DSP and ML algorithms.

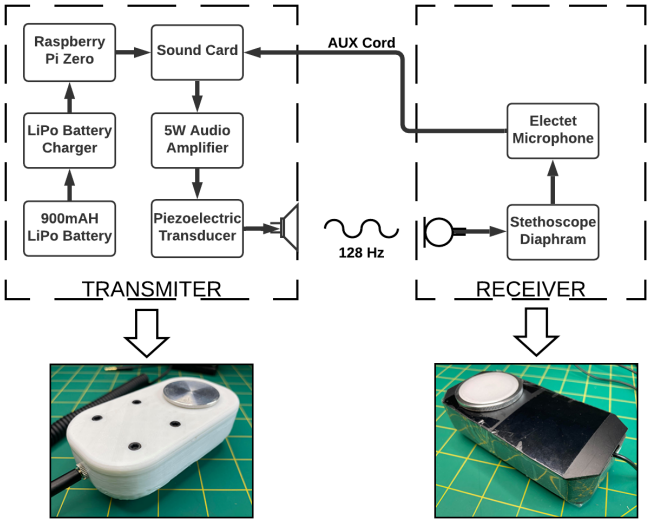

Fig. 1. Block diagram and current iteration of prototype hardware.

In order to maximize adaptability and minimize cost, the hardware is designed using commercial components controlled by a Pi-Zero microcomputer (Fig. 1). A piezoelectric transducer transmits a 128 Hz sine wave at 5 W (although an arbitrary signal could be generated) through the layer of flesh into the underlying bone structure and acoustic signals are recorded using a compact microphone connected to a stethoscope diaphragm to improve the sensitivity when measuring small vibrations through thicker layers of soft tissue, both electrically mimicking the methods used in previous work [9]–[11]. Digital to analog and analog to digital conversion was achieved using a commercially available Pi-zero sound-card. The device does not require internet connectivity as data collection and processing are completed with the device's microcomputer.

Data collection was acquired using bovine, porcine, and ovine femurs, chosen for their similarity in structure and size to human femurs [16]. Femurs were purchased from local butcher shops. Half of the femurs were fractured and the others were left whole. Femurs were fractured by applying strain using a vice or impact using a hammer to the body of the femur, resulting in non-displaced transverse, oblique, and comminuted fractures. In order to simulate soft-tissue surrounding bones, the femurs used to collect data were encased in gelatin (Fig. 2), which is often used as a stand-in for flesh in biomedical tests [12]. The gel was trimmed to between 1 cm and 5 cm thickness perpendicular to the bone. Gel was trimmed less near the center-point of the femurs, where flesh would be thickest. Gel was trimmed closer near the femur 'knuckles' (i.e., larger rounded ends of the femur), where bones are often covered in a thin layer of flesh (e.g., near the knees and elbows). Five equally spaced marks were made on the surface of the gel using a pen. The first and last marks were made where the femur's 'knuckles' were of largest diameter.

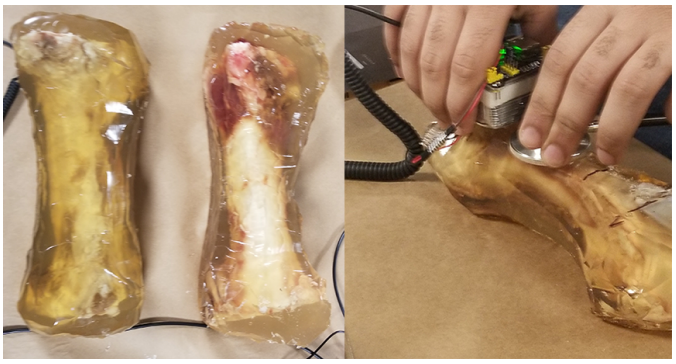

Fig. 2. Ovine bone training pair in ballistic gel simulating soft tissue (left) and data collection being performed at various collection positions marked in black (right).

Data from each bone was collected using a prototype device including a piezoelectric transducer and microphone. Thirty or sixty second clips were recorded at a variety of transducer and microphone positions (i.e., marks on gel) on each bone.

Data was collected on two bovine, two porcine, and six ovine femurs for a total of 60 minutes of audio during several sessions. Three femurs were used for holdout-testing and seven femurs were used for training, with balanced class distribution of broken and intact bones. After collecting stream of audio, 100 millisecond windows were extracted.

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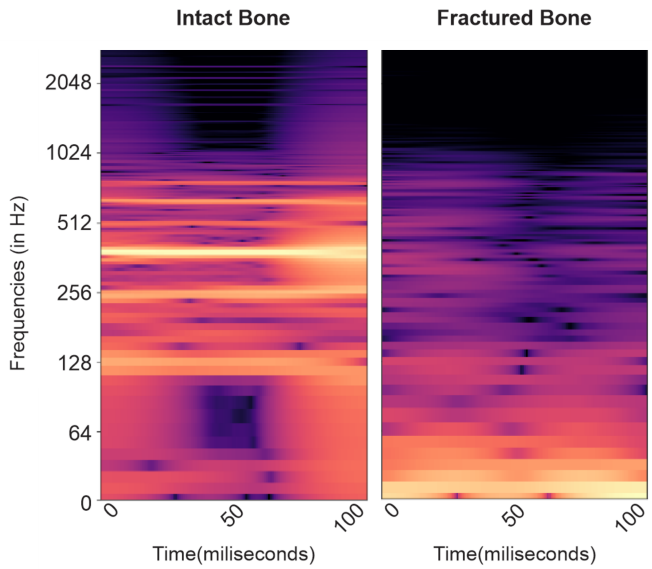

Fig. 3. Power spectrum showing significant harmonics at integer multiples of 128 Hz (256 Hz, 384 Hz, 512 Hz,...) in non-fractured control bone (left) in comparison to those seen in fractured bone (right).

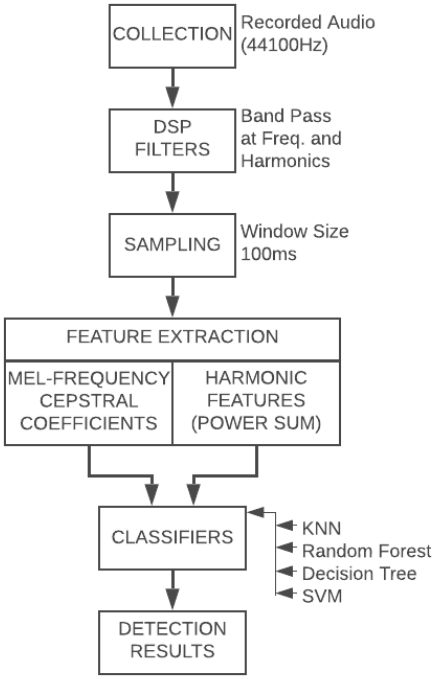

Fig. 4. Methodology of the digital signal processing and machine learning pipeline.

Inducing frequency signals in bones can produce harmonics at each integer multiplication of that frequency (i.e., a 128 Hz signal will produce harmonics at 256 Hz, 384 Hz…). However, the amplitude of the harmonic typically diminishes as the frequency increases, making it hard to distinguish between noise and signal at higher harmonic frequencies. Given these results, five narrow bands of 10 Hz around the first five harmonics were extracted from the signal and used as training data. A fifth order Butterworth bandpass filter was used to extract these narrow bands and merge them together. The overall data-processing pipeline is described in Figure 4.

Two methods were used to extract features from the dataset: MFCC (Mel Frequency Cepstrum Coefficients) and features based on observed characteristics of harmonics and power.

The power of a harmonic decreases as the harmonic frequency increases, thus lower frequencies are more important. MFCC weights lower frequencies higher as compared to higher frequencies, similar to how humans perceive sound; therefore, it was utilized as a feature.

Additionally, features based on the observation that harmonics were not as prominent in fractured bones as whole bones were extracted. Specifically, the ratio of the signal strength (power at 128 Hz) to the sum of the harmonic power, the signal power, the difference between the sum of the power of the harmonics and the signal power, and the ratio of the signal power to each of the harmonics individually were calculated.

IV. Results

A number of machine learning classifiers were compared during research. The dataset was created by combining both sets of features discussed above (MFCC and harmonic features). Initially both feature sets were analyzed individually, achieving 85% accuracy using MFCC features and about 90% using harmonic features. The algorithm's accuracy was increased to 93.6% by combining both sets of features. Specifically, Support Vector Machine (SVM), Random Forest, Decision Tree, and K Nearest Neighbors (KNN) with varied parameters were compared. Support vector machines performed best, yielding a total accuracy of 93.6% and an F1 score of 95.4% (Table I).

TABLE I
Test Set Results: Harmonic Features

Algorithm	***Accuracy***	***F1 Score***
RBF SVM	93.6	95.4
Random Forest	89.9	92.25
Decision Tree	84.97	84.51
KNN	83.56	85.60

The combined MFCC and harmonic feature-set was too highly dimensional to visualize (dimension size=186). As a result, t-SNE was applied in order to compress the dataset into two dimensions so that class clusters could be easily visualized. As shown in Figure 5, classes formed clear clusters with clear boundaries. The algorithm segmented audio into 100 ms clips with an 70 ms stride and classified clips correctly 93.6% of the time.

V. Discussion

The fracture detection method researched by Moore [13], which relies on trained doctors and a stethoscope, can be implemented using an automated embedded system. Digital

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Fig. 5. t-SNE plot to visualize the clustering between classes. (Blue dots: Intact Bones; Orange dots: Fractured Bones)

signal processing techniques and machine learning algorithms can be used to quickly and reliably differentiate between fractured and whole bones using cheap, widely available hardware. While further testing is needed to see if these results on simulated fractures will hold for the detection of in-vivo animal and human fractures, the results strongly suggest that a low-cost diagnostic device is possible.

The proof-of-concept algorithm presented here was built from limited training and test sets comprised of bones encased in simulated soft tissue; however, more research is required to confirm that this system could be used for fracture detection in humans or animals. For example, a rigorous in-vivo study may further evidence the presented technique's generalizability. Future research should also focus on generating a larger data set in order to increase the potential for differentiation between fracture types.

Beyond long-bone fracture detection, future work may extend the presented technique to additional use-cases including assessment of spinal injuries by first responders, wrist or ankle injuries by medical clinicians, and general bone injuries by military field-medics and sports medicine practitioners. In addition to human applications, fracture detection in pets and livestock could prove valuable, as animals cannot communicate where pain is localized.

VI. Conclusion

The presented method—which incorporates digital signal processing and machine learning—in combination with the proof-of-concept device, offers a low-cost, easy-to-use fracture detection solution that can accurately distinguish between fractured and whole bones.

Prior research indicates that acoustic signals are a viable tool for fracture detection in bone. The findings presented here suggest previous methods of acoustic bone fracture detection can be automated and improved using DSP and ML in a similar way to industrial acoustic fracture detection techniques proposed in the literature. While not a replacement for radiographic diagnosis, this technique and device offer an avenue for further research.

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